Conditioned approximation for minimax sampling and estimation

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Abstract

Sampling and estimation must be chosen for an estimand and a risk criterion. We study exact-size sampling and design-unbiased linear estimation under bounded structured residual uncertainty and positive diagonal noise. A natural marginal relaxation collapses to the classical noise-allocation floor and can miss arbitrarily large structured risk. Conditioning on membership of a selected set restores useful information. We identify a convex primal for the conditioned certificate and recover an actual sampling strategy by conditional dependent rounding. Its risk is at most $1 + 2\Lambda_B$ times the joint optimum, where $\Lambda_B = \max_{i \notin B} \rho^2 \|F_i\|^2/d_i$; at most $2^{|B|}$ convex blocks are required. The construction is exact when structure is supported on the conditioned set. Analytic examples establish sharpness and show why strong units need not be certainty selections. Exact optimization is NP-hard even with one feature. With equal noise-weighted target contributions, standard cube balancing already supplies near optimality in a complementary large-sample regime. The proposed contribution is the conditioned joint-strategy guarantee; minimax design, optimal allocation, perspectives, and dependent rounding are established ingredients.

1 Introduction

A sampling rule should be chosen for the quantity to be estimated and the loss attached to its error. We construct a sampling strategy that jointly chooses inclusion probabilities, dependence, and sample-dependent unbiased linear weights. The method conditions on membership of a selected set of units and solves a convex program over the resulting cells. It then draws an exact-size sample by dependent rounding.

The guarantee depends on the structured uncertainty left outside the conditioned set. If its largest structured-to-noise row ratio is Λ_B, the strategy from an optimal relaxed solution has risk at most $1 + 2\Lambda_B$ times the full joint minimax optimum. Structure supported on the conditioned set gives exact optimality. The number of convex blocks is at most $2^{|B|}$, so the method is useful when a small conditioning set leaves a sufficiently weak tail. A numerical implementation supplies a feasible strategy and risk bounds checked in exact arithmetic without the optimizer, even when the uniform factor is conservative.

The contribution is this constructive guarantee and its conditions of usefulness. Optimal allocation, minimax design, conic perspectives, and dependent rounding are established ingredients. With equal noise-weighted target contributions, ordinary balancing already gives near optimality in a complementary regime; the analytic examples below do not establish practical superiority over optimized existing methods.

2 Decision problem

Let the finite population be $\{1, \dots, N\}$, and let $\mathcal{A}$ be a finite nonempty family of allowed samples. A sampling strategy chooses probabilities q_s on $\mathcal{A}$ and estimates the linear target $T_a(y) = a^T y$. The vector a, a prediction m_0, a feature matrix $F \in \mathbb{R}^{N \times d}$, positive diagonal matrix $D = \operatorname{diag}(d_i)$, and radius $\rho \geq 0$ are known before collection. Residual uncertainty is

$$y - m_0 = F\beta + \varepsilon, \qquad \|\beta\|_2 \leq \rho, \quad E\varepsilon = 0, \quad \operatorname{Cov}(\varepsilon) = D \succ 0. \tag{1}$$

Sampling is independent of ε. Risk integrates over sampling and noise and maximizes over β. The estimator class consists of

$$\widehat{T} = a^T m_0 + w_s^T (y - m_0), \qquad \operatorname{supp}(w_s) \subseteq s, \qquad \sum_{s \in \mathcal{A}} q_s w_s = a. \tag{2}$$

Weights can depend on the sample identity and precollection information, but not on observed residual values. The last equality is design unbiasedness for every population. Assume **target coverage**: every coordinate with $a_i \neq 0$ belongs to at least one allowed sample. This ensures existence of a finite-risk strategy. The class includes Horvitz–Thompson (HT) weights and permits more general sample-dependent, possibly negative weights. It excludes nonlinear and outcome-dependent estimators.

Writing $Q = \sum_s q_s w_s w_s^T - aa^T$, direct expansion gives

$$R(q, w) = \operatorname{tr}(DQ) + \rho^2 \lambda_{\max}(F^T Q F), \qquad R^* = \min_{q,w} R(q, w). \tag{3}$$

When F has no columns, omit the structured term in (3).

The estimand enters through a, and the uncertainty class enters through F, D, ρ. Both determine the sampling decision. Structured minimax experimental design already uses this decision-theoretic logic; equal treatment marginals can make the independent-noise term constant, whereas free sampling marginals make allocation part of the problem. This is a change in the admissible decisions, not a new principle of optimality. (Kallus, 2020, Section 2)

Joint sampling and estimation also has direct survey predecessors, including classical minimax survey analysis. The present mixed-risk problem is not asserted to be theorem-equivalent to those formulations. (Cheng and Li, 1983) For known bounded outcome intervals and an expected sample-size budget, a recent result optimizes over all design-unbiased estimators: independent inclusion with probabilities proportional to interval widths, capped at one, and midpoint-differenced HT estimation attain minimax risk. Its rectangular outcome class and expected-size constraint differ from (1) and an exact-size family. This nearby solved problem precludes treating joint optimization itself as the contribution. (Aronow and Lopatto, 2026, Theorems 1–3)

Let $P = \operatorname{conv}\{\mathbf{1}_s : s \in \mathcal{A}\}$ be the exact marginal polytope and define

$$B_P = \min_{p \in P} \sum_i a_i^2 d_i (1/p_i - 1). \tag{4}$$

A zero-target term is zero even at $p_i = 0$; a nonzero-target term at $p_i = 0$ is infinite. Coverage makes the minimum finite. For any strategy (2), Cauchy–Schwarz gives $E w_{S,i}^2 \geq a_i^2 / p_i$. The structured risk is nonnegative, so $R^* \geq B_P$. Conversely, when $\rho = 0$, any noise-optimal $p \in P$ can be realized by a design, and its HT estimator attains (4). Thus B_P is exactly the noise-only optimum in the same feasible family.

The diagonal-noise floor is the weighted-target form of the classical Godambe–Joshi anticipated-variance bound. Nedyalkova and Tillé derive the design-unbiased linear bound, its clipped optimal

allocation, and attainment by exactly balanced HT under their linear model. Applying their total-estimation formula to $a_i(y_i - m_{0i})$ gives the variance term $a_i^2 d_i$ on nonzero target coordinates. The arbitrary feasible-family minimum in (4) is the formulation used here, rather than a claim that every such generalization appears in the classical source. (Godambe and Joshi, 1965; Nedyalkova and Tille, 2008, equation (2.9), Result 2.2, and Strategy 2.1) The equation and strategy numbers refer to the reprint, not the journal pagination. Exact balance is the attainment condition; approximate landing needs a separate error bound.

For all size-n samples, $P = \{p \in [0,1]^N : \mathbf{1}^T p = n\}$. If more than n target coefficients are nonzero, the clipped allocation $p_i = \min\{1, c|a_i|\sqrt{d_i}\}$, with c chosen to fill the sample size, minimizes (4). Otherwise set every target inclusion probability to one and fill any remaining capacity with zero-target units; then $B_P = 0$. A noise-optimal allocation need not minimize (3). The distinction is whether dependence and weighting can suppress the structured term while retaining those marginals.

3 Method and constructive guarantee

We now restrict the allowed family to all size-n samples, or disjoint exact stratum quotas, retaining the model and estimator class (1)–(3). Write $B_0 = B_P$ for the corresponding quota-constrained noise floor. No constructive assertion is made here for arbitrary overlapping constraints.

Fix a set B of b units. Partition allowed samples by their intersection with B, retaining only nonempty cells $\mathcal{A}_j$. Let $P_j = \mathrm{conv}\{1_s : s \in \mathcal{A}_j\}$. Each cell fixes the head indicators to zero or one and has an exact residual tail quota. The construction does not require $b < n$.

3.1 A convex primal equal to the conditioned certificate

For each cell choose a mass $q_j \geq 0$, an unconditional inclusion vector $t_j \in q_j P_j$, and an unconditional weight contribution $x_j \in \mathbb{R}^N$. Impose

$$\sum_j q_j = 1, \qquad \sum_j x_j = a.$$

Use closed perspectives: $x^2/t = 0$ at $x = t = 0$, and infinity at $t = 0, x \neq 0$. Define

$$V = \sum_{j,i} d_i \frac{x_{ji}^2}{t_{ji}} - a^T D a,$$
$$W = \sum_j \frac{(F^T x_j)(F^T x_j)^T}{q_j} - (F^T a)(F^T a)^T.$$

and minimize

$$\Phi(q, t, x) = V + \rho^2 \lambda_{\max}(W). \tag{5}$$

At zero mass, finiteness forces $t_j = x_j = 0$; its matrix perspective is zero. For positive mass put $p_j = t_j/q_j$ and $k_j = x_j/q_j$. Then $W = \mathrm{Cov}_j(F^T k_j) \succeq 0$. Also

$$V = \underbrace{\sum_j q_j \sum_i d_i k_{ji}^2 (1/p_{ji} - 1)}_{V_{\text{within}} \geq 0} + \underbrace{\sum_i d_i \{\sum_j q_j k_{ji}^2 - a_i^2\}}_{V_{\text{between}} \geq 0}. \tag{6}$$

Terms with $p_{ji} = 0$ are zero because $k_{ji} = 0$. This is a convex conic program: scalar quadratic perspectives represent the noise terms and matrix perspectives represent the structured second moments. There are at most 2^b cell blocks, each polynomial in N, d.

Proposition 1. *Write $L_{\mathcal{C}}$ for the minimum of (5). It equals the dual certificate (40), derived in the appendix. It is attained and satisfies $0 \le L_{\mathcal{C}} \le R^*$.*

Proof. Aggregate any actual design by its cell. Set $x_j = E[w1_{J=j}]$ and $t_{ji} = P(J = j, i \in S)$. Coordinatewise Cauchy–Schwarz gives $E[w_i^2 1_{J=j}] \ge x_{ji}^2/t_{ji}$. Conditional Jensen gives $E[(F^T w)(F^T w)^T 1_{J=j}] \succeq (F^T x_j)(F^T x_j)^T/q_j$. Thus (5) is a relaxation. Nonnegativity follows from (6) and $W \succeq 0$.

For completeness, its dual recovers the stated conditioned certificate. Dualize the spectral epigraph constraint with a matrix multiplier $H \succeq 0, \operatorname{tr} H = 1$, set $U = \rho F H^{1/2}$, and use $2z$ as the multiplier for $\sum x_j = a$. For fixed $p_j \in P_j$, minimizing over k_j gives the negative of

$$\begin{aligned} J_j(H, z, p_j) &= \sup_k \{2z^T k - \sum_i d_i k_i^2/p_{ji} - \|U^T k\|^2\} \\ &= \min_v \{\|v\|^2 + \sum_i p_{ji}(z_i - u_i^T v)^2/d_i\}. \end{aligned}$$

Optimizing p_j and applying convex-concave minimax (compact P_j) yields $G_j(H, z) = \min_v \max_{p \in P_j} \{\|v\|^2 + \sum_i p_i(z_i - u_i^T v)^2/d_i\}$. Minimizing the cell masses yields $-\max_j G_j$. The remaining terms give $2a^T z - G_{\mathcal{C}}(H, z) - a^T(D + UU^T)a$, as required.

There is no duality gap: choose all cell masses positive, inclusions in each cell's relative interior, and supported x_j summing to a, using target coverage. After deleting identically zero coordinates, take strict slack in all perspective epigraphs and the spectral upper-bound constraint. This is relative Slater feasibility. Positive (6), $V \ge 0$, and $W \succeq 0$ bound the weight perspectives on objective sublevels; compact masses and inclusions and closed perspectives give attainment. Omit the spectral variables when $\rho = 0$ or there are no feature columns. □

3.2 An actual sampler and a relative-risk guarantee

Given any finite feasible point of (5):

1. Draw a cell $J = j$ with probability q_j.
2. Keep its fixed indicators and use mean-preserving pairwise rounding on the remaining p_j, separately within each stratum if applicable. This preserves exact quotas and gives nonpositive off-diagonal conditional indicator covariances.
3. Use weights $w_i = I_i k_{ji}/p_{ji} = I_i x_{ji}/t_{ji}$, with zero weights when $p_{ji} = 0$.

The cell is determined by the observed sample's intersection with B, so these are sample-dependent weights in the original class, not a larger class requiring unrecorded estimator randomization. Conditional expectation is $E[w \mid J = j] = k_j$; hence $Ew = a$. Conditional diagonal second moments are k_{ji}^2/p_{ji}, so its noise risk is exactly V.

Define the computable tail strength

$$\Lambda_B = \max_{i \notin B} \frac{\rho^2 \|F_i\|^2}{d_i},$$

with zero for an empty tail. It is fixed by the stated uncertainty model, not fitted using sampled outcomes.

Theorem 2. *(constructive approximation).*
The preceding sampler satisfies

$$R_{\text{round}} \le \Phi + \Delta \le \Phi + 2\Lambda_B V_{\text{within}} \le (1 + 2\Lambda_B)\Phi, \tag{7}$$

where a sharper computable additive allowance is

$$\Delta = 2\rho^2 \sum_{j:q_j>0} q_j \sum_{i:0<p_{ji}<1} \frac{k_{ji}^2(1-p_{ji})}{p_{ji}} \|F_i\|^2. \tag{8}$$

In particular, an optimal relaxed solution gives

$$L_{\mathcal{C}} \le R^* \le R_{\text{round}} \le (1 + 2\Lambda_B)L_{\mathcal{C}} \le (1 + 2\Lambda_B)R^*. \tag{9}$$

An exactly feasible relaxed solution with $\Phi \le L_{\mathcal{C}} + \delta$ instead gives $R_{\text{round}} \le (1 + 2\Lambda_B)(R^* + \delta)$.

Proof. Let $C_j = \operatorname{Cov}(I \mid j)$. Pairwise rounding gives $C_{j,i\ell} \le 0$ off the diagonal, and fixed sample size gives $C_j 1 = 0$. Therefore

$$C_j \preceq 2\operatorname{diag}(p_{ji}(1-p_{ji})).$$

Indeed, for every vector v, $v^T C_j v = \sum_{i<\ell}(-C_{j,i\ell})(v_i - v_\ell)^2 \le 2\sum_i p_{ji}(1-p_{ji})v_i^2$. Fixed head indicators have zero conditional covariance. By total covariance,

$$F^T Q F = W + \sum_j q_j F^T \operatorname{diag}(k_j/p_j) C_j \operatorname{diag}(k_j/p_j) F.$$

Every added matrix is positive semidefinite. Its spectral contribution is at most its trace, which is bounded by (8). Since head terms vanish, $\Delta \le 2\Lambda_B V_{\text{within}} \le 2\Lambda_B V \le 2\Lambda_B \Phi$. Add the exact noise risk. This proves all assertions. □

This proof controls the *additional within-cell covariance* against *within-cell noise risk*. It does not use the large offset $a^T D a + \rho^2 \|F^T a\|^2$ from the value-only argument in the appendix. It needs no rank bound, no lower inclusion-probability bound, no $b < n$, and no condition $\Lambda_B < 1$. A large factor is valid but may be uninformative.

Sharpness of the certificate factor. With $B = \varnothing, N = 2, n = 1, a = (1,1), D = I, \rho = 1, F = (t, -t)^T$, the two-unit calculation in the appendix gives $L_{\mathcal{C}} = 2$ and $R^* = 2 + 4t^2$. Since $\Lambda_B = t^2$, the factor $1 + 2\Lambda_B$ in the integrality-gap inequality is attained. This establishes sharpness of a uniform comparison to this relaxation. It does not show that the rounded sampler's ratio to R^* is sharp; on this instance that sampler is optimal.

Sharpness for unrestricted pairwise rounding. Take $N = 4, n = 2, a = \mathbf{1}, D = I, \rho = 1, F = (t, -t, t, -t)^T$, and no conditioning. The relaxed optimizer has $p_i = 1/2, k_i = 1$ and $\Phi = 4$. Pairwise rounding the pairs $\{1,2\}$ and $\{3,4\}$ independently selects one unit from each. If the independent signs are ϵ_1, ϵ_2, then $F^T w = 2t(\epsilon_1 + \epsilon_2)$, giving structured risk $8t^2$ and total risk $4 + 8t^2$. But choosing $\{1,2\}$ or $\{3,4\}$ with equal probabilities and weights two attains $R^* = B_0 = 4$. Thus an allowed pairwise rounding implementation can attain the full ratio $1 + 2\Lambda_B$. The reference implementation uses these pairs on this ordering of the rows; its within-cell rounding ignores the features. Reordering the feature rows to $(t, t, -t, -t)^T$ instead makes its rounding attain risk 4, with the same certificate upper bound $4 + 8t^2$. This does not preclude a feature-aware choice of rounding pairs, or cube balancing, from improving the sampler.

Tolerance rule. To obtain a factor $1 + \eta$ at an exact relaxed optimum, take $B \supseteq \{i : \rho^2 \|F_i\|^2 / d_i > \eta/2\}$. If the chosen conditioning set has b members, at most 2^b blocks suffice. For an

approximate solve, take $B \supseteq \{i : \rho^2\|F_i\|^2/d_i > \eta/4\}$, so $2\Lambda_B \le \eta/2$ and, when the classical floor $B_0 > 0$, certify $\delta \le \eta B_0/(2+\eta)$. Then $(1+\eta/2)(R^* + \delta) \le (1+\eta)R^*$. This is a theorem about a finite conic formulation and its certified approximation error; a full finite-bit complexity and numerical-certification theorem is not supplied.

When $B_0 = 0$, every nonzero target coordinate fits in the sample budget; include all those coordinates and use $w = a$, attaining risk zero exactly. Under disjoint quotas, the corresponding quota-constrained zero floor has the same implication stratum by stratum. This handles the zero-risk case without dividing by a risk lower bound.

A posteriori stopping rule. With a feasible relaxed point, compute $U_R = \Phi + \Delta$. With any certified conditioned-dual lower value L, put $L_0 = \max\{B_0, L\}$. Then $L_0 \le R^* \le R_{\text{round}} \le U_R$. If $U_R \le (1+\eta)L_0$, the constructed sampler is certified within the desired factor even when the uniform tail bound is large. Here Φ is a primal *upper* value for the relaxation and must not be mistaken for a lower bound on R^*.

4 When conditioning is useful

If $F_i = 0$ outside B, Theorem 2 is exact: the relaxed optimum recovers an optimal strategy. The selection decisions on the head remain randomized and optimized; conditioning does not force the head into every sample. The formulation can have exponentially fewer blocks than the full sample list when $b \ll N$.

A formal separation from two simple comparators uses $N = 3, n = 2, D = I, \rho = 1$,

$$a = (T^{-2}, 1, 1)^T, \qquad F = (T, 0, 0)^T, \qquad T \ge 2.$$

Condition on unit one. There are two cells and a zero-strength tail, so the constructive relaxation is exact. Censusing unit one leaves one of units two and three to sample and has optimum risk exactly two, by the tail noise floor and its HT attainment.

In contrast, choose samples $\{1,2\}, \{1,3\}, \{2,3\}$ with probabilities $1/(2T), 1/(2T), 1 - 1/T$, and use ordinary HT. Its inclusion probabilities are $\pi_1 = 1/T$, $\pi_2 = \pi_3 = 1 - 1/(2T)$, and its risk is

$$R_{\text{HT}} = \frac{1+T^2}{T^4}(T-1) + \frac{2}{2T-1} < \frac{3}{T}.$$

Thus $R^* < 3/T$ and the census comparator's ratio diverges. Noise-optimal marginals have $\pi_1 = 2T^{-2}/(2+T^{-2})$, $\pi_2 = \pi_3 = 2/(2+T^{-2})$. Any noise-floor-attaining strategy must use HT weights, and its structured risk alone equals $1 - 1/(2T^2)$, also an unbounded ratio relative to R^*. This example separates from censusing the strong unit and blindly optimizing noise alone. It does not establish superiority over an optimized cube design or all existing sampling algorithms; the displayed HT construction itself is a simple successful comparator. This one-row example is itself solved by classical allocation after replacing d_1 by $1+T^2$: with $h = \sqrt{1+T^2}/T^2$, the exact optimum is $R^* = 2h - h^2/2$, attained by HT with marginals proportional to $(h, 1, 1)$. Thus the example isolates the failure of the two stated comparators; it does not show that the new optimization is needed for this instance.

4.1 A complementary guarantee from established balancing

A constructive comparator is cube flight followed by mean-preserving pairwise landing. Flight preserves auxiliary totals and leaves only dimension-many unresolved inclusion indicators. Pairwise landing preserves exact sample size and conditional marginals. These are established components of

balanced sampling and dependent rounding. The risk inequalities below are deductions from those components, not claims that the displayed formulas were previously published verbatim. (Deville and Tillé, 2004; Chauvet, 2009, equations (5)–(7); Gandhi et al., 2006, Theorem 2.3)

Proposition 3. *(a direct fixed-size risk bound).*

Allow all size-n samples, $1 \le n < N$, and choose $\pi \in [0,1]^N$ summing to n, positive on every nonzero target coordinate. Let $r = \operatorname{rank}(F)$, and set

$$b(\pi) = \sum_i a_i^2 d_i (1/\pi_i - 1), \qquad A(\pi) = \max_{i: a_i \neq 0} \frac{\rho^2 a_i^2 \|F_i\|^2}{\pi_i^2}, \tag{10}$$

where the maximum is zero for the zero target. There exists an exact-size, design-unbiased HT strategy, constructed by cube flight and pairwise landing, such that

$$R_{\text{cube}}(\pi) \le b(\pi) + \frac{r+1}{2} A(\pi). \tag{11}$$

Proof. Write $X_{\pi,i} = a_i F_i / \pi_i$, with zero rows for zero-target units. Start flight at π and preserve $\mathbf{1}^T p = n$ and $X_\pi^T p = X_\pi^T \pi$. A nullspace step among fractional coordinates, choosing its endpoints with probabilities that preserve the current mean, fixes at least one coordinate. At termination its random endpoint p satisfies

$$Ep = \pi, \qquad X_\pi^T (p - \pi) = 0, \qquad \#\{i : 0 < p_i < 1\} \le r + 1. \tag{12}$$

The count follows because more than $r+1$ fractional columns admit a nonzero nullspace direction for the size and feature constraints. Integral coordinates remain fixed.

Round the remaining entries using mean-preserving pairwise steps. Each step changes two coordinates in opposite directions; integer sample size ensures that a nonintegral state has at least two fractional entries. Conditional on the flight endpoint, the final indicator S has $E(S \mid p) = p$ and constant size. Its covariance $C = \operatorname{Cov}(S \mid p)$ has nonpositive off-diagonal entries: martingale increments are orthogonal across steps, and each step has nonpositive off-diagonal covariance. Also $C\mathbf{1} = 0$ and $C_{ii} = p_i(1 - p_i)$. Consequently,

$$x^T C x = \sum_{i<j} (-C_{ij})(x_i - x_j)^2 \le 2 \sum_i p_i (1 - p_i) x_i^2. \tag{13}$$

Flight eliminated feature imbalance, so the HT structured risk is bounded by

$$\begin{aligned} \rho^2 \lambda_{\max}\Big(E[X_\pi^T (S - \pi)(S - \pi)^T X_\pi] \Big) &\le \rho^2 E \|X_\pi^T (S - \pi)\|^2 \\ &\le 2\rho^2 E \sum_i p_i (1 - p_i) \|X_{\pi,i}\|^2 \\ &\le \frac{r+1}{2} A(\pi). \end{aligned} \tag{14}$$

The last step uses (12) and $p_i(1 - p_i) \le 1/4$. Marginal preservation gives $ES = \pi$, and hence exact HT noise risk $b(\pi)$, independently of sampling dependence. Adding the terms proves (11). □

Minimizing the right-hand side of (11) over the cardinality polytope is a convex allocation problem: $1/\pi_i$ and $1/\pi_i^2$ are convex on positive coordinates, and a maximum of convex functions is convex. This gives a simple allocation-and-balance comparator using the same precollection information as the joint problem. The maximum-row penalty can be loose, particularly near a census, and (11) alone supplies no sharp ratio for arbitrary heterogeneous inputs.

For the family of all size-n samples, the resulting bounds give the explicit sandwich

$$B_0 \le R^* \le \inf_{\substack{\pi\in[0,1]^N,\ \mathbf{1}^T\pi=n \\ \pi_i>0 \text{ if } a_i\neq 0}} \left\{ b(\pi) + \frac{r+1}{2} A(\pi) \right\}. \tag{15}$$

Thus collapse to the noise floor can be practically harmless when the balanced upper bound is close to that floor. Equation (15) uses the unrestricted cardinality family; it does not assert existence of the same cube construction for an arbitrary $\mathcal{A}$.

Initial certainty units never move in flight or landing. For a fixed π, they can therefore be omitted from the maximum in (10). To preserve a convex allocation comparison, preselect a fixed set C, $|C| = c < n$, require $\pi_i = 1$ on C, and optimize the remaining probabilities with budget $n - c$. The same proof gives

$$R_{\mathrm{cube},C}(\pi) \le \sum_{i\notin C} a_i^2 d_i (1/\pi_i - 1) + \frac{r+1}{2} \max_{\substack{i\notin C \\ a_i\neq 0}} \frac{\rho^2 a_i^2 \|F_i\|^2}{\pi_i^2}. \tag{16}$$

The maximum is zero when no nonzero target remains outside C. For each fixed C, (16) is a convex optimization over the tail probabilities. If the maximum instead excludes whichever units the optimizer happens to set to one, the objective generally loses convexity: at a certainty boundary a previously positive maximum term can disappear. Selection of C is a separate discrete decision. Censusing a small exceptional set is consequently a relevant comparator to conditioning on its membership; neither procedure uniformly dominates the other from these bounds.

Corollary 4. *(equal noise-weighted target contributions).*
Suppose $|a_i|\sqrt{d_i} = k > 0$ for all i, and $\rho^2\|F_i\|^2/d_i \le \Lambda$. Uniform marginals $\pi_i = f = n/N$ are noise-optimal and give

$$B_0 = Nk^2(1-f)/f, \qquad \boxed{\frac{R_{\mathrm{cube}}}{R^*} \le 1 + \frac{(r+1)\Lambda}{2n(1-n/N)}.} \tag{17}$$

Proof. In (11), $A(\pi) \le k^2\Lambda/f^2$. Divide by $B_0 \le R^*$. □

No condition $F^T a = 0$ or $(r+1)\Lambda/(2n) < 1$ is needed. In this regime, (17) and $B_0 \le R^*$ also give $B_0 \le R^* \le R_{\mathrm{cube}} \le B_0\{1 + (r+1)\Lambda/[2n(1-n/N)]\}$. The noise-floor certificate is already nearly tight when the displayed excess is small.

Simple random sampling alone is an insufficient benchmark here. With $Z = \mathrm{diag}(a)F$ and $P_0 = I - \mathbf{1}\mathbf{1}^T/N$, its exact risk under these uniform marginals is

$$R_{\mathrm{SRS}} = B_0 + \rho^2 \frac{1-f}{f} \frac{N}{N-1} \lambda_{\max}(Z^T P_0 Z). \tag{18}$$

The resulting ratio upper bound is $1 + N\Lambda/(N-1)$, and an attaining example shows that this difference is substantive. Take even $N > n$, even n, $a_i = d_i = 1$, $\rho > 0$, and a single feature with half the rows $+\sqrt{\Lambda}/\rho$ and half $-\sqrt{\Lambda}/\rho$. Selecting $n/2$ units uniformly from each group preserves uniform marginals and balances the estimand-weighted feature rows $Z_i = a_i F_i$, so it attains $R^* = B_0$. Since $Z^T P_0 Z = N\Lambda/\rho^2$, (18) gives exactly

$$\frac{R_{\mathrm{SRS}}}{R^*} = 1 + \frac{N\Lambda}{N-1}. \tag{19}$$

An upper-bound comparison alone would not establish this example's separation. It follows here from an explicit feasible design attaining the universal lower bound. Modern Gram–Schmidt-walk and thinning results provide additional comparators under their admissible size constraints. (Harshaw et al., 2022, Theorem 4.1; Carrell et al., 2025)

5 Computable certificates and implementation

The reference implementation solves the conditional conic program with CVXPY and Clarabel. It scales the coordinates by their noise standard deviations and centers the perspective terms to avoid cancellation. Numerical outputs propose a strategy and dual witnesses; the numerical optimizer's termination status does not certify them.

Cell masses, conditional inclusion probabilities, and weight contributions are repaired in rational arithmetic. The repair preserves exact quotas and restores unbiasedness. When a joint inclusion probability decreases, its contribution is reduced proportionately before the unbiasedness correction, preventing the probability repair from amplifying the proposed sample weight. Zero-target coordinates remain available as control variates.

A verifier checks feasibility, the complete family of feasible head patterns, and both risk witnesses without calling the optimizer. It shares arithmetic routines with certificate construction; the stored witnesses also permit third-party checking. The upper bound uses exact noise and covariance calculations and a rational positive-semidefiniteness check of a spectral upper witness. The lower bound uses a repaired trace-one positive definite adversary and exact cellwise score maximization; an independently valid noise witness remains available if dual recovery fails. Bounds refer to the binary64 input values interpreted exactly, not to uncertainty about the chosen model.

The optimizer minimizes the relaxed objective Φ, rather than the final risk upper bound $\Phi + \Delta$. With no conditioning, the sampler's actual risk can attain the full $1 + 2\Lambda_B$ factor in the sharpness example above, so poor performance need not be an artifact of a loose certificate.

Relaxation convergence means that its certified gap meets the requested absolute and relative tolerances. Near optimality of the sampler requires the separate comparison of its risk upper bound with a positive lower bound on the full optimum. A small absolute tolerance can dominate at tiny risk scales; a purely relative request sets the absolute tolerance to zero. Failure to reach requested accuracy retains the feasible strategy and valid bounds in the returned exception.

Sampling uses rational Bernoulli probabilities and pairwise rounding within each stratum. The mathematical guarantee applies to the exact weighted estimator under ideal randomness; conversion of its value to floating point introduces ordinary rounding error. Closed-form optima, exhaustive branching of small sampling trees, unit rescaling, corrupted witnesses, and injected numerical failures provide regression checks. These checks establish implementation behavior on specified cases, not comparative statistical performance or a finite-bit complexity theorem.

6 Relationship to prior work

6.1 The closest prior results establish the ingredients

Kallus's minimax design analysis supplies the spectral objective and the need for randomized mixtures. Classical survey theory supplies allocation bounds and balanced attainment. Sagnol's multiresponse optimal-design formulation supplies the conic design-and-weight perspective. Cube flight, pairwise rounding, and the fixed-sum covariance inequality supply the rounding tools. Disaggregation over selected binary choices is standard convex optimization. These facts preclude broad claims of a new sampling philosophy or a new convexification principle.

Talovic and Tillé's robust-design analysis is particularly close. Their fixed-size covariance bound and design-mixture construction directly overlap the ingredients used here. Their mixture preserves prescribed first-order probabilities; its exact covariance decomposition averages conditional covariance over the first design. Those results do not by themselves optimize our inclusion probabilities and

sample-dependent weights or establish Theorem 2's ratio. (Talovic and Tillé, 2025, thesis Chapter 2, Corollary 3, Algorithm 1 and Result 11)

Joint optimization itself also has recent exact solutions under other uncertainty classes and collection constraints, including the rectangular-outcome, expected-budget result cited above. The comparison must therefore be at the level of this conditioned guarantee, not at the level of optimizing a sampling design for a loss.

6.2 Conditional Optimal Sampling has a different optimization theorem

The supplementary proof of García-Segador and Miranda's Theorem 3 fixes inclusion probabilities and an estimator depending on the design only through them. Expected loss is then linear in sample probabilities and attains its minimum at a vertex. (García-Segador and Miranda, 2026, supplementary proof, page 1)

That conclusion does not extend to the minimax risk even on a shared fixed-marginal HT subclass. Take $N = 4, n = 2, a = \mathbf{1}, D = I_4, F = I_4, \rho > 0$, prescribe $\pi_i = 1/2$, and use $w = 2\mathbf{1}_S$. The marginal equations force $p_{12} = p_{34}$, $p_{13} = p_{24}$, and $p_{14} = p_{23}$. The feasible polytope is therefore the simplex of mixtures of three complementary-pair designs, with masses $\lambda_1, \lambda_2, \lambda_3$. Write

$$v_1 = (1, 1, -1, -1)^T, \quad v_2 = (1, -1, 1, -1)^T,$$
$$v_3 = (1, -1, -1, 1)^T.$$

These vectors are orthogonal with squared norm four, giving

$$Q = \sum_{j=1}^{3} \lambda_j v_j v_j^T, \qquad R = 4 + 4\rho^2 \max_j \lambda_j.$$

The unique minimizing mixture is $\lambda_j = 1/3$, namely simple random sampling on all six samples, with risk $4 + 4\rho^2/3$. Every vertex has risk $4 + 4\rho^2$. Maximizing over outcome directions has changed a linear loss into a convex spectral objective whose optimum is strictly interior. This illustrates Kallus's mixed-strategy distinction: restricting the minimax optimization to these vertices would be invalid. The inspected COS vertex argument does not directly deliver the joint guarantee; an additional minimax construction and risk bound are needed. It is not a claim that every result of that article has been checked; the comparison uses its fully accessible supplementary proof.

Ordinary samplewise calibration can eliminate structured error but does not by itself ensure design unbiasedness. Any regression-adjusted comparator must satisfy $Ew = a$ to belong to the benchmark class used here. Exact balance together with noise-optimal marginals already attains the universal noise floor.

7 Limits and implications

Conditioning gives a finite convex representation and a constructive risk guarantee under exact size or disjoint quotas. It can require exponentially many cells, and exact rational verification can be expensive. The reference implementation is intended for modest conditioning sets and feature dimensions. No polynomial finite-bit solver complexity or practical dominance over optimized cube or Gram–Schmidt Walk procedures is established.

The result covers linear estimands, a fixed precollection uncertainty model, and supported design-unbiased linear weights. Overlapping collection constraints, learned uncertainty sets, nonlinear estimands, nonresponse, and general variance estimation require additional analysis. The literature

comparison supports a specific conditioned approximation contribution, not a claim that joint optimization or minimax sampling is new.

A Exact dual and marginal relaxation

A.1 The exact dual requires a global subset bound

For $d > 0$, introduce $H \succeq 0$, $\operatorname{tr} H = 1$, and

$$M(H) = D + \rho^2 F H F^T.$$

If $\rho = 0$ or there are no feature columns, omit H and use $M = D$. For any vector z, define

$$g_s(H, z) = z_s^T M(H)_{ss}^{-1} z_s, \qquad g^*(H, z) = \max_{s \in \mathcal{A}} g_s(H, z), \tag{20}$$

with score zero for the empty sample.

Proposition 5. *(exact finite benchmark).*
Under (1)–(2), positive diagonal noise, and target coverage,

$$R^* = \max_{H,z} \{2a^T z - g^*(H, z) - a^T M(H) a\}. \tag{21}$$

Both the primal minimum and dual maximum are attained. Consequently, any proved upper bound $G \geq g^*(H, z)$ gives

$$2a^T z - G - a^T M a \leq R^*, \qquad \frac{(a^T z)^2}{G} - a^T M a \leq R^* \quad (G > 0). \tag{22}$$

The second expression optimizes the scale of z. A feasible strategy supplies an upper bound on R^*, giving a valid risk interval.

Proof. Lift $\xi_s = q_s w_s$ and impose

$$\begin{pmatrix} V_s & \xi_s \\ \xi_s^T & q_s \end{pmatrix} \succeq 0, \quad \sum_s \xi_s = a, \quad \operatorname{supp}(\xi_s) \subseteq s, \quad q \geq 0, \quad \sum_s q_s = 1. \tag{23}$$

Set $Q_V = \sum_s V_s - a a^T$, constrain $F^T Q_V F \preceq tI$, and minimize $\operatorname{tr}(D Q_V) + \rho^2 t$. Omit t, its constraint, and its objective term when $\rho = 0$ or F has no columns. At $q_s = 0$, the block constraint forces $\xi_s = 0$; set $w_s = 0$ there. Otherwise it gives $V_s \succeq \xi_s \xi_s^T / q_s$, and recover $w_s = \xi_s / q_s$. Replacing every V_s by this exact perspective, or zero when $q_s = 0$, recovers supported unbiased weights without increasing either risk term. The lift is therefore exact.

Summing (23) gives $Q_V \succeq 0$. Positive D bounds Q_V, each positive-semidefinite V_s, and the ξ_s on objective sublevels. The simplex bounds q; t is bounded whenever retained. Closedness gives primal attainment. Coverage permits positive q_s on every sample and supported ξ_s summing to a. Taking $V_s = \xi_s \xi_s^T / q_s + cI$, $c > 0$, and sufficiently large t when retained gives strict conic feasibility relative to the equalities. Strong duality follows. The samplewise dual minimization is $\inf_{\operatorname{supp}(w) \subseteq s} \{w^T M w - 2 z^T w\} = -g_s(H, z)$. The simplex constraint therefore produces $\gamma \geq g_s(H, z)$ for every sample; eliminating γ gives (21). Equivalently, complete the square on each sample:

$$w_s^T M w_s \geq 2 z^T w_s - g_s(H, z).$$

Averaging and using unbiasedness proves the first bound in (22). At scale t, the bound is $2ta^Tz - t^2G - a^TMa$; maximizing this quadratic proves the second. □

For fixed H, let E_s embed sample coordinates and put $\mathcal{I}(q) = \sum_s q_s E_s M_{ss}^{-1} E_s^T$. Optimizing supported unbiased weights gives $a^T\mathcal{I}(q)^+a - a^TMa$, provided a lies in the range of $\mathcal{I}(q)$. This is multiresponse c-optimal design, so the perspective construction has established optimal-design antecedents. Its sample family can contain $\binom{N}{n}$ members; exact finite convexity is not a polynomial representation. Additional restrictions on the probabilities, such as expected costs or prescribed marginals, require their own dual terms. (Sagnol, 2010, Theorems 3.3 and 4.3)

Fix H, and factor $M = D + UU^T$, where row i of U is u_i^T. Completing a ridge square gives

$$g_s(H,z) = \min_v \left\{ \|v\|^2 + \sum_{i\in s} \frac{(z_i - u_i^Tv)^2}{d_i} \right\}. \tag{24}$$

The exchange of maximum and minimum therefore yields the upper bound

$$g^*(H,z) \le \min_v G_z(v), \qquad G_z(v) = \|v\|^2 + \max_{p\in P} \sum_i p_i \frac{(z_i - u_i^Tv)^2}{d_i}. \tag{25}$$

For exact sample size, the maximum sums the n largest residual scores; for disjoint stratum quotas, it sums the appropriate largest scores in each stratum. The function is strongly convex and coercive in the feature-factor dimension. Every evaluated v gives a valid upper bound, without requiring convergence of its minimization.

This is the standard diagonal-perspective relaxation of sparse quadratic optimization. Indeed, its optimized value also equals

$$\max_{p\in P,x} \left\{ 2z^Tx - \sum_i d_i x_i^2/p_i - \|U^Tx\|^2 \right\}, \tag{26}$$

with the closed perspective at zero coordinates. Perspective and Boolean relaxations of this kind precede the sampling formulation. (Askari et al., 2021, Sections 2.1–2.3; Pilanci et al., 2015, Sections 2–3)

Theorem 6. *(noise-floor collapse).*

For every finite allowed sample family with target coverage, every factor U, and the exact marginal polytope P,

$$\boxed{\sup_{z,v}\{2a^Tz - G_z(v) - a^T(D+UU^T)a\} = B_P.} \tag{27}$$

Every scaled certificate in (22) obtained from $G_z(v) > 0$ is also at most B_P. Optimizing v exactly or maximizing over H does not change this conclusion.

Proof. Put $w = z - Uv$ and

$$T_P(w) = \max_{p\in P} \sum_i p_i w_i^2/d_i.$$

Since $a^TMa = a^TDa + \|U^Ta\|^2$, square completion yields

$$2a^Tz - G_z(v) - a^TMa = 2a^Tw - T_P(w) - a^TDa - \|v - U^Ta\|^2. \tag{28}$$

For fixed v, the substitution between z and w is bijective. The supremum over v removes the final square. The function $2a^T w - \sum_i p_i w_i^2/d_i$ is continuous, affine in p, and concave in w. Compactness and convexity of P permit convex-concave minimax interchange, giving

$$\begin{aligned}\sup_w\{2a^T w - T_P(w)\} &= \min_{p\in P}\sup_w\left\{2a^T w - \sum_i p_i w_i^2/d_i\right\}\\ &= \min_{p\in P}\sum_i a_i^2 d_i/p_i.\end{aligned} \tag{29}$$

The original payoff is finite and continuous even when some $p_i = 0$; only its evaluated supremum acquires the boundary values in (29). The compactness required by the interchange is supplied by P, not by a truncation of w or exclusion of zero probabilities. Subtracting $a^T Da$ proves (27). Scaling cannot escape the family: $G_{tz}(tv) = t^2 G_z(v)$, so each scalar rescaling is another value covered by (28). Finally, $\sup_{z,v}$ already allows exact minimization of $G_z(v)$ for every z, and the conclusion holds for each H separately. □

The exact marginal polytope is insufficient for this certificate. The theorem applies even when every inclusion constraint induced by the sample family is retained. Replacing P by a looser polytope can weaken the noise bound further, but recovering P exactly cannot restore the structured contribution in (27). The limitation is specific to this unconditioned ridge/perspective construction. It is not an impossibility theorem for other convex relaxations or for sampling optimization.

The information interpretation identifies the lost constraint more precisely. For strictly positive p, define

$$J(p) = \left[\operatorname{diag}(d_i/p_i) + UU^T\right]^{-1}.$$

Its continuous boundary extension satisfies $J(\mathbf{1}_s) = E_s M_{ss}^{-1} E_s^T$. For every z, the ridge identity expresses $z^T J(p) z$ as the infimum over v of a function affine in p, so J is matrix concave. Therefore $J(E_q \mathbf{1}_S) \succeq E_q J(\mathbf{1}_S) = \mathcal{I}(q)$. The relaxation replaces the actual information of a mixture of samples by this larger fractional information. It inflates only the diagonal of $J(p)^{-1}$, retaining the full UU^T; the structured term consequently cancels when $a^T Ma$ is subtracted. This is a concave extension, not a claim that it is the exact concave envelope. Other functions of the marginals could encode the best attainable risk at those marginals and need not suffer this failure.

For all size-n samples, write $B_0 = B_P$. Combining the scaled ridge certificate with the noise floor gives exactly, whenever $G_z(v) > 0$,

$$\max\left\{0, B_0, \frac{(a^T z)^2}{G_z(v)} - a^T Ma\right\} = B_0. \tag{30}$$

More accurate ridge optimization cannot improve this lower endpoint. A design algorithm can still improve the feasible upper risk, but that is a different achievement. With stratum quotas, collapse occurs at the quota-constrained noise floor; an improvement over an unconstrained allocation bound reflects those quotas, not structured uncertainty.

A.2 A two-unit instance separates the bound from the actual optimum

Take $N = 2$, $n = 1$, $a = (1, 1)^T$, $D = I$, $\rho = 1$, and $F = (t, -t)^T$, with $t > 0$. If unit one is selected with probability p, design unbiasedness forces singleton weights $1/p$ and $1/(1-p)$. Thus the general linear class equals the HT class here, and

$$R(p) = (1 + t^2)\{1/p + 1/(1-p)\} - 2, \qquad R^* = 2 + 4t^2, \qquad B_0 = 2. \tag{31}$$

The optimum uses $p = 1/2$. Its covariance is $Q = \left(\begin{smallmatrix} 1 & -1 \\ -1 & 1 \end{smallmatrix}\right)$, with noise risk two and structured risk $4t^2$. Although $F^T a = 0$, the structured component remains a nuisance in the sampled observation. Orthogonality of the target and the feature does not make estimation costless.

For $z = (1, 1)^T$, both sample scores equal $1/(1 + t^2)$, while

$$G_z(v) = v^2 + (1 + t|v|)^2, \qquad \min_v G_z(v) = 1. \tag{32}$$

Both feasible subsets are already optimal for the true score. The discrepancy is entirely in the upper relaxation. Equation (27) nevertheless limits its optimized risk certificate to two, while $R^*/B_0 = 1 + 2t^2$ diverges. This example rules out fixing the certificate merely by improving the subset rounding rule.

B Conditioned dual and supporting pricing bounds

Conditioning restricts fractional information to a union of cell polytopes with fixed inclusion decisions, rather than maximizing it over their full convex hull. The ridge representation expresses that restriction through separate node variables. This removes some of the information inflation identified above; it is not a general prohibition against methods expressed through marginals.

Proposition 7. *(membership-conditioned bounds).*

Fix H, z, and write the expression inside braces in (24) as $f_s(v)$. Partition $\mathcal{A}$ into nonempty cells $\mathcal{A}_j$ according to membership of a chosen set of b units. Define

$$G_{\mathcal{C}} = \max_j \min_v \max_{s \in \mathcal{A}_j} f_s(v). \tag{33}$$

$$g^*(H, z) \le G_{\mathcal{C}} \le \min_v \max_{s \in \mathcal{A}} f_s(v). \tag{34}$$

Refining the partition cannot increase $G_{\mathcal{C}}$. If every cell contains one sample, (33) equals the exact score maximum.

Proof. Each cell obeys the max-min inequality in (25), so its optimized ridge bound dominates every sample score in that cell. Taking the maximum gives the first inequality. For any common v, each cell minimum is at most its value at v; maximizing across cells and minimizing over this common vector gives the second inequality. Each child cell is a subfamily of its parent, so its bound is at most the parent's bound. Singleton cells are exact by (24). □

For a cardinality constraint, a cell's additive maximization includes the forced-in scores, excludes forced-out scores, and selects the largest remaining scores needed to fill the residual quota. Disjoint strata use their residual quotas separately. Every cell is therefore another low-dimensional convex problem. At most 2^b cells occur, and infeasible membership patterns are discarded. Exactness can require exponentially many cells. This is standard branching, not a new computational principle or a polynomial-time guarantee.

Approximate cell solves remain valid: evaluate any v_j in each cell and maximize the resulting upper bounds. Monotonicity of these raw evaluations does not follow automatically from refinement. A child can inherit its parent's evaluated vector and retain the smaller of that valid bound and a new evaluated bound, ensuring monotonicity without assuming convergence. Arithmetic errors still require certified bounds if a rigorous floating-point interval is claimed.

The conditioning repair is strict in (31). Conditioning on inclusion of unit one leaves two singleton cells. At $z = (1, 1)^T$, each optimized cell bound is $1/(1 + t^2)$. The scaled lower certificate is therefore

$$\frac{(a^T z)^2}{G_{\mathcal{C}}} - a^T M a = 4(1+t^2) - 2 = 2 + 4t^2 = R^*. \tag{35}$$

One inclusion decision restores all missing structured risk in this analytical example. The result establishes a discriminating test for another relaxation: it should certify some positive part of $R^* - B_P$ on an instance where that premium is unavoidable. Success on the two-unit test does not establish usefulness for large frames.

B.1 A small exceptional set gives a quantitative pricing guarantee

Theorem 8. *(conditioning with a controlled tail).*

Allow all size-n samples. Fix H, z, use a factor U with at most r columns, and choose a set B of $b < n$ units such that

$$\|u_i\|^2/d_i \le \Lambda_{\text{tail}} \quad (i \notin B), \qquad \epsilon_B = \frac{(r+1)\Lambda_{\text{tail}}}{2(n-b)} < 1. \tag{36}$$

There is no restriction on the feature magnitudes inside B. Solve the convex ridge problem in each feasible membership cell, sparsify its exact saddle point, and round the remaining fractional coordinates. The largest returned score satisfies

$$g_{\widehat{s}} \ge (1-\epsilon_B) G_{\mathcal{C}} \ge (1-\epsilon_B) g^*(H, z). \tag{37}$$

The construction uses at most 2^b convex node problems. The condition $\rho^2\|F_i\|^2/d_i \le \Lambda_{\text{tail}}$ outside B makes the guarantee uniform over all admissible H.

Proof. A cell fixes the selected subset $A \subseteq B$, excludes $B \setminus A$, and leaves free quota $m = n - |A| \ge n - b > 0$ in B^c. Its objective is

$$G_A(v) = \|v\|^2 + \sum_{i \in A} \ell_i(v) + \text{Top}_m\{\ell_i(v) : i \notin B\}, \qquad \ell_i(v) = (z_i - u_i^T v)^2/d_i.$$

At an exact saddle point, the free fractional vector p has sum m and satisfies $v = \sum_{i\in A} b_i(v) + \sum_{i \notin B} p_i b_i(v)$, where $b_i(v) = u_i(z_i - u_i^T v)/d_i$. Preserving these r equations and the active face leaves at most $r+1$ fractional free coordinates. The fixed contribution from A changes the right-hand side of the stationarity equations, not their number.

Use the common inverse upper bound $W_A = (I + \sum_{i\in A} u_i u_i^T/d_i)^{-1} \preceq I$. Every completion contains A. Fractional free coordinates can remain only when $0 < m < N - b$. Let τ be the mth largest free score. Exchange optimality on the active maximizing face forces every fractional coordinate to tie at τ, so $G_A^* \ge m\tau$. The rounding calculation in the appendix, with zero score and stationarity residuals, gives

$$G_A^* - g_{s_A} \le \frac{r+1}{2}\Lambda_{\text{tail}}\tau \le \frac{(r+1)\Lambda_{\text{tail}}}{2m} G_A^* \le \epsilon_B G_A^*.$$

An integral saddle point is exact, and $\tau = 0$ gives zero rounding error directly. Apply this inequality to a cell attaining $G_{\mathcal{C}} = \max_A G_A^*$, and choose the best returned sample over all cells. This proves (37). Finally, $H \preceq I$ implies $\|u_i\|^2 = \rho^2 F_i H F_i^T \le \rho^2\|F_i\|^2$, proving uniformity. □

Approximate node solutions use the full score-gap, stationarity, and rounding errors rather than assuming (37). If node A has evaluated upper bound $\widetilde{G}_A$, returned score g_{s_A}, and proved error $E_A \ge \widetilde{G}_A - g_{s_A}$, then

$$0 \le \max_A \widetilde{G}_A - \max_A g_{s_A} \le \max_A E_A. \tag{38}$$

This follows by considering a node attaining the largest upper bound. The explicit expression for E_A is the difference between its upper bound and the right-hand side of the deterministic version of (45), including all fixed coordinates. Thus inexact convex solves need not be treated as exact saddle points.

There is a sharper zero-tail case. If $u_i = 0$ for every $i \notin B$, the requirement $b < n$ is unnecessary. Each feasible cell has exact optimized score

$$z_A^T (D_A + U_A U_A^T)^{-1} z_A + \mathrm{Top}_{n-|A|}\{z_i^2/d_i : i \notin B\}. \tag{39}$$

The selected covariance is block diagonal between A and the free units, proving the formula. Maximizing over at most 2^b cells gives exact pricing; cells with infeasible residual quotas are discarded. The choice $F_i = 0$ outside B suffices uniformly over adversaries.

Theorem 8 supplies a cost/accuracy statement for a small number of strong rows and a weak tail. It is a consequence of conditioning and the established rounding bound, not a new branching principle. As b approaches n, the conservative factor in (36) degrades even though the optimized bound (33) improves monotonically. The next result gives the corresponding minimax-value guarantee and retains the estimand-dependent subtraction explicitly.

B.2 Conditioning gives a convex minimax-value certificate

Corollary 9. *(conditioned value approximation).*

Suppose a fixed membership partition satisfies, uniformly over H, z, $g^*(H,z) \le G_{\mathcal{C}}(H,z) \le g^*(H,z)/\alpha$, where $0 \le \epsilon < 1$ and $\alpha = 1 - \epsilon$. Theorem 8 supplies $\epsilon = \epsilon_B$ for the family of all size-n samples, conditioning on $b < n$ units, with $r \ge \mathrm{rank}(F)$ and the uniform bound $\rho^2 \|F_i\|^2/d_i \le \Lambda_{\mathrm{tail}}$ outside that set. Other families require their own uniform factor; the cardinality factor is not asserted for partition quotas. Define

$$\begin{aligned} L_{\mathcal{C}} &= \sup_{H,z}\{2a^T z - G_{\mathcal{C}}(H,z) - a^T M(H) a\}, \\ C_a &= a^T D a + \rho^2 \|F^T a\|^2. \end{aligned} \tag{40}$$

Then

$$\boxed{\max\{B_P, R^* - \epsilon(R^* + C_a)\} \le L_{\mathcal{C}} \le R^*.} \tag{41}$$

In particular, a certified optimization bracket $L_- \le L_{\mathcal{C}} \le L_+$ yields

$$\boxed{\max\{B_P, L_-\} \le R^* \le \frac{L_+ + \epsilon C_a}{1-\epsilon}.} \tag{42}$$

This is a bound on the minimax value in (3), not the risk of a constructed estimator.

Proof. Weak duality gives $L_{\mathcal{C}} \le R^*$. Conditioning decreases the unconditioned pricing bound, so Theorem 6 gives $L_{\mathcal{C}} \ge B_P$. At an exact maximizer (H^*, z^*) of (21), put $\gamma = g^*(H^*, z^*)$ and $c^* = a^T M(H^*) a$. Scalar stationarity of the quadratic objective along tz^* gives $a^T z^* = \gamma$ and $R^* = \gamma - c^*$. For a nonzero target, $\gamma = R^* + c^* > 0$, since $R^* \ge 0$ and $c^* \ge a^T D a > 0$. Homogeneity gives $G_{\mathcal{C}}(H, tz) = t^2 G_{\mathcal{C}}(H,z)$, so evaluation at αz^* gives

$$\begin{aligned} L_{\mathcal{C}} &\geq 2\alpha\gamma - \alpha^2 G_{\mathcal{C}}(H^*, z^*) - c^* \\ &\geq \alpha\gamma - c^* = R^* - \epsilon(R^* + c^*). \end{aligned}$$

Every admissible $H \preceq I$, so $c^* \leq C_a$, proving (41). For $a = 0$, both R^* and $L_{\mathcal{C}}$ are zero. Rearranging $(1-\epsilon)R^* \leq L_{\mathcal{C}} + \epsilon C_a$, then using the certified upper endpoint L_+, proves (42). □

A feasible point of the maximization in (40) provides only a lower bound on $L_{\mathcal{C}}$. Substituting that value for L_+ in (42) is invalid; the upper endpoint needs a certified optimization upper bound, for example an appropriate conic dual bound. A feasible sampling strategy with risk U_R can independently tighten the right endpoint to the smaller of U_R and (42). Proposition 1 identifies a primal of (40) from which Theorem 2 constructs such a strategy.

The optimization in (40) is convex. Let $\Omega = \rho^2 H$. Replacing a node's ridge variable by $h = \Omega^{1/2} v$ gives its optimized bound as

$$G_j(H, z) = \inf_h \left\{ h^T \Omega^+ h + \max_{s \in \mathcal{A}_j} \sum_{i \in s} (z_i - F_i h)^2 / d_i \right\},$$

where the matrix perspective is infinite unless $h \in \operatorname{range}(\Omega)$. The perspective and the residual squares are jointly convex in their variables; partial minimization and the maximum over nodes preserve convexity. Thus $G_{\mathcal{C}}$ is jointly convex in H, z, and the objective in (40) is concave.

For cardinality or disjoint partition-quota cells, an explicit conic formulation uses separate h_j, t_j in each node. Let A_j be its forced-in set, R_j its free set, and m_j its residual quota. Introduce residual upper bounds ℓ_{ij}, free-score threshold θ_j, nonnegative excesses e_{ij}, and one common bound T. Impose

$$\begin{aligned} &\begin{pmatrix} \rho^2 H & h_j \\ h_j^T & t_j \end{pmatrix} \succeq 0, \qquad d_i \ell_{ij} \geq (z_i - F_i h_j)^2, \\ &e_{ij} \geq \ell_{ij} - \theta_j, \quad e_{ij} \geq 0 \quad (i \in R_j), \\ &T \geq t_j + \sum_{i \in A_j} \ell_{ij} + m_j \theta_j + \sum_{i \in R_j} e_{ij}. \end{aligned} \tag{43}$$

The block constraint enforces the range condition as well as the quadratic perspective. Residual constraints are rotated second-order-cone constraints. Maximizing $2a^T z - T - a^T D a - \rho^2 (F^T a)^T H (F^T a)$ subject to (43) and $H \succeq 0, \operatorname{tr} H = 1$ represents (40). For partition quotas, use one threshold and excess sum per stratum. Zero or full residual quotas can be written directly. When $\rho = 0$ or F has no columns, omit H, h_j, t_j and the structured term, leaving the ordinary diagonal scores z_i^2/d_i. No inversion of a singular H is needed.

For $\rho > 0$ and $d > 0$, choose $H = I/d$, $h_j = 0$, $t_j > 0$, and strict slack in the residual, excess, and T inequalities. This gives relative strict conic feasibility; finite value then gives strong conic duality. It supports arbitrarily tight upper optimization bounds in exact mathematics, not automatic certification from floating-point residuals.

The number of node blocks is at most 2^b. This finite representation does not establish a finite-bit complexity bound or numerical certification from ordinary residual tolerances. The constructive result in Theorem 2 supplies a strategy without propagating the offset in (41). The value error is informative when ϵC_a is small relative to the risk scale. In particular, (41) guarantees a certificate strictly above B_P whenever $R^* - B_P > \epsilon(R^* + C_a)$. If the target loads strongly on a structured direction, C_a can be large even when the achievable risk is small; the value approximation can then be weak despite accurate pricing.

C Hardness of exact optimization

Proposition 10. *Deciding whether $R^* = B_0$ for the full design-unbiased linear class is NP-hard, already for exact half-sampling, $a = \mathbf{1}, D = I, \rho = 1$, and a single nonnegative integer feature column.*

Proof. Given positive integers $c_1, \dots, c_m$ in PARTITION, take $N = 2m, n = m$, and $F = (c_1, \dots, c_m, 0, \dots, 0)^T$. The noise floor is $B_0 = N$. Equality in the strictly convex marginal allocation bound forces every $\pi_i = 1/2$. Equality in coordinatewise Cauchy–Schwarz then forces $w_i = 2I_i$ almost surely. Equality in total risk additionally requires

$$2\sum_{i\in S} F_i = \sum_i F_i$$

on every supported sample. Such a sample supplies a solution to PARTITION. Conversely, a partition of the original numbers can be padded with zero-feature units to size m. Assign probability one half each to that sample and its complement. It has marginals one half, uses weights $2I_i$, and exactly balances F, attaining N. The transformation has polynomial bit length. Attainment follows from Proposition 1 and Theorem 2 with all units conditioned, where the cells are singleton samples and the rounding loss is zero. Thus equality at the floor cannot be an unattained infimum. □

This is hardness of the statistical optimization itself, not merely of subset pricing. The reduction uses weakly NP-complete PARTITION. It does not prove strong NP-hardness, a constant-factor inapproximability barrier, or the impossibility of an FPTAS. It also gives an obstruction to multiplicatively approximating the *excess* $R^* - B_0$: a finite factor would need to detect its exact zero. That observation concerns an exact guarantee and does not establish a positive additive-gap hardness bound.

D Supporting quadratic rounding bound

Fix $M = D + UU^T$, z, v, and feasible fractional marginals p under an exact size or disjoint quota constraint. Put

$$\ell_i = (z_i - u_i^T v)^2/d_i, \quad b_i = u_i(z_i - u_i^T v)/d_i, \quad C_s = I + \sum_{i\in s} u_i u_i^T/d_i.$$

$$f_s(v) - g_s = \left(v - \sum_{i\in s} b_i\right)^T C_s^{-1} \left(v - \sum_{i\in s} b_i\right). \tag{44}$$

Equation (44) follows by expanding (24) around its minimum.

Let $W \succeq C_s^{-1}$ for all rounded completions. The choice $W = I$ always works; if $A = \{i : p_i = 1\}$, the sharper choice $W = (I + \sum_{i\in A} u_i u_i^T/d_i)^{-1}$ is valid because every completion contains A. Conditional negatively correlated rounding has mean p and covariance satisfying (13). With $e = v - \sum_i p_i b_i$, (44) gives

$$Eg_S \geq \|v\|^2 + \sum_i p_i \ell_i - e^T W e - 2\sum_i p_i(1-p_i)\|b_i\|_W^2. \tag{45}$$

Consequently some feasible sample attains this bound. A deterministic construction uses

$$\Psi(x) = \sum_i x_i \ell_i - \left\| v - \sum_i x_i b_i \right\|_W^2 - 2\sum_i x_i(1-x_i)\|b_i\|_W^2.$$

Along a quota-preserving swap $x_i \mapsto x_i + t$, $x_j \mapsto x_j - t$, its quadratic coefficient is $\|b_i + b_j\|_W^2 \geq 0$. Choosing the better endpoint preserves or increases Ψ, proving the deterministic version of (45). This is a specialization of deterministic pipage rounding. (Ageev and Sviridenko, 2004; Harvey and Olver, 2013, Theorem 7.2)

At an exact ridge saddle point, preserve its active maximizing face and r stationarity equations while eliminating fractional coordinates. If more fractional coordinates remain than the number of independent equality rows, a nonzero nullspace move fixes another coordinate at an endpoint. The remaining count is therefore at most $r + 1$ for a uniform quota. For partition quotas, f fractional coordinates in h fractional strata satisfy $f \leq r + h$ and $f \geq 2h$, hence $f \leq 2r$. Approximate nonactive inputs preserve their linear score as one additional equality. The counts become $f \leq r + 2$ for a uniform quota and $f \leq r + h + 1$, $f \geq 2h$, for partition quotas, giving $f \leq 2(r + 1)$. These are elementary extreme-point counts of the same kind used in cube flight.

The factor two in (45) is sharp for this generic diagonal variance bound when a valid exact common inverse is used. In (32), the saddle point has $p = (1/2, 1/2)$, $v = 0$, $b_1 = -b_2 = t$, and exact common inverse $W = 1/(1 + t^2)$. The loss in (45) equals $t^2/(1 + t^2) = 1 - g_s$, exactly the relaxation gap. The anchored choice based on initially selected units is instead $W = I$ in this example. The calculation therefore does not prove sharpness of the constant in Theorem 8.

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